

the problem, the Memory Manager under the kernel can now dynamically manage the address space—allocating and de-allocating resources as the workload demands. Thus, the amount of virtual memory used to store paged buffers can grow when device drivers ask for more, and it can shrink when the drivers release it.

3.7 Advanced Features

Windows Vista comes packed with some forward-looking technologies. For example, it can make use of external Flash drives and portable music players as additional memory to boost the performance of a system. Vista also offers an advanced caching system which caches frequently used programs to memory such that future launches of these programs are significantly faster. While these features are internal to the OS itself, Vista also offers an innovation in the form of Windows SideShow which will help take personal Windows data and relevant information to external devices such as phones and laptops. Let's take a look at some of these new and advanced features of Vista.

3.7.1 Memory Priorities And SuperFetch

Here's a fairly common scenario under older versions of Windows: let's say you were working on some vital data, say a particularly complex worksheet under Excel, and decided to take a lunch break. While gone, let's assume that an anti-virus program is triggered to scan your hard drive as part of a screen-saver, or even as a scheduled task. When the anti-virus is triggered, it might request the system to give access to more memory than is currently available. Under this scenario, the system will flush all data associated with your Excel work to the hard drive and free up memory for the anti-virus program. Now, when you return to your Desktop and try to resume your work, you are likely to notice performance degradation: Excel will take a long time to restore to full screen, for example, while the entire system gets busy freeing the necessary space in memory